

Lab 4: Architecture & UML for Your Client

Formative (ungraded) · Moodle-enabled · output is the first half of Task 4 (15%, due Week 11)

Item	Details
Linked to	Week 5 lecture — Designing Software Architecture with UML (ULO2, ULO4)
Work mode	Individually, on the SAME client you chose for Task 2
Time	About 90 minutes
Tools	diagrams.net desktop OR PlantUML — both free, both work offline, both cached on Moodle
Submission	Diagrams + this worksheet to Moodle AND committed to week05/ in your Git portfolio

Do this properly and Task 4 gets much easier. Task 4 (Design & Modelling Deliverable, due Week 11) asks for an outline architecture plus UML models. The two diagrams you produce today are exactly that architecture section. Weeks 9–10 add the behavioural diagrams.

Part A — Answer the six design questions (25 min)

Answer for *YOUR* client (Kina Ridge Primary School or Waghi Valley Produce Cooperative). One or two sentences each — but each answer must be a decision, not a description.

#	Question	Your decision, and the requirement that forced it
1	What are the major parts of the system?	
2	How do the parts communicate?	
3	Where does the data live?	
4	Where does each part physically run?	
5	How is access controlled, and where is that enforced?	
6	What happens when it fails — power loss, no signal, duplicate sync?	

Part B — Choose an architectural pattern (10 min)

Pattern chosen (or blend): _____

In 100–150 words: why this pattern for THIS client? Name one pattern you rejected and the specific requirement that ruled it out.

Warning: if you chose client–server with one central database, check it against your Part A question 6 answer. At a site with no reliable connectivity that choice fails on day one.

Part C — Draw the package diagram (25 min)

Show your system as 3–5 logical packages with dependency arrows. Rules: arrows point downward only; every package has one clear responsibility; no package is called “Miscellaneous”.

If you are using PlantUML, start from this

```
@startuml
package "UI" {
    [Fee Entry Form]
}
package "Application Logic" {
    [Fee Service]
}
package "Data Access" {
    [Payment Repository]
}
[Fee Entry Form] --> [Fee Service]
[Fee Service] --> [Payment Repository]
@enduml
```

Save as **week05/architecture-packages.puml** (or **.drawio** from diagrams.net).

Part D — Draw the deployment diagram (25 min)

Show every device your system runs on, what is installed on each, and the link between them. This is the diagram that exposes PNG delivery risk, so be honest about the network.

Must appear on your diagram	Check
Every physical device (laptop, phone, office PC, printer)	
What software component sits on each device	
Where the data is stored on each device	
The link between devices, labelled with how and how often it is available	
A note on what still works while that link is DOWN	

PlantUML deployment starter

```
@startuml
node "Depot laptop" {
    [Intake App]
    database "Local SQLite"
}
node "Town office PC" {
    [Consolidation]
    database "Master DB"
}
[Intake App] ..> [Consolidation] : USB or 4G when available
@enduml
```

Part E — Commit (5 min)

1. Put both diagram files and this worksheet into week05/ in your portfolio.
2. Commit with a message that says what you decided, not just what you added:

```
git add week05/  
git commit -m "Add offline-first architecture: package and deployment diagrams"
```

AI use: you may use AI to draft PlantUML text. You must then check it against your own Part A answers — generated architectures default to cloud client–server and routinely ignore power loss and duplicate sync. Declare AI use in your README.